

Introduction to Artificial Intelligence

IICT Lecture 09

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Lecturer

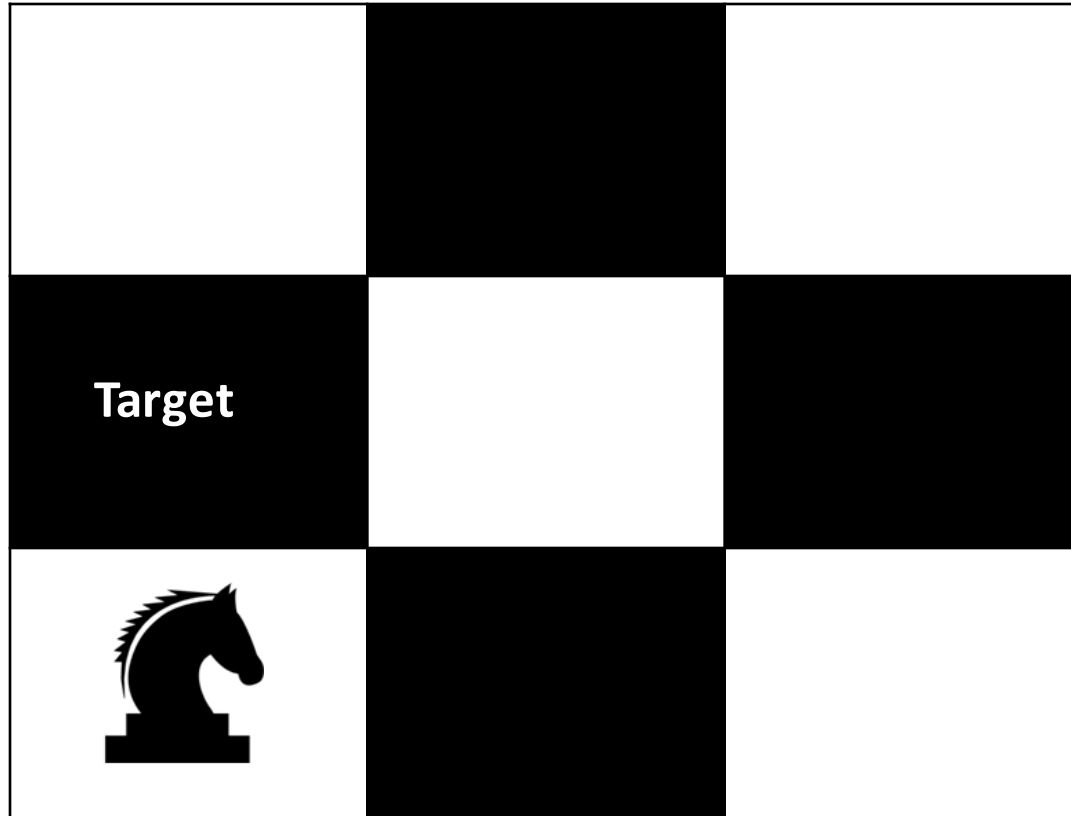
FAST NU CS Dept.

Intelligence

- A very general mental capability that involves the ability to **reason**, **plan**, **solve problems**, **think abstractly**, **comprehend complex ideas**, **learn quickly** and **learn from experience**. [1]
- Measurement of Intelligence:
 - Intelligence Quotient (IQ)

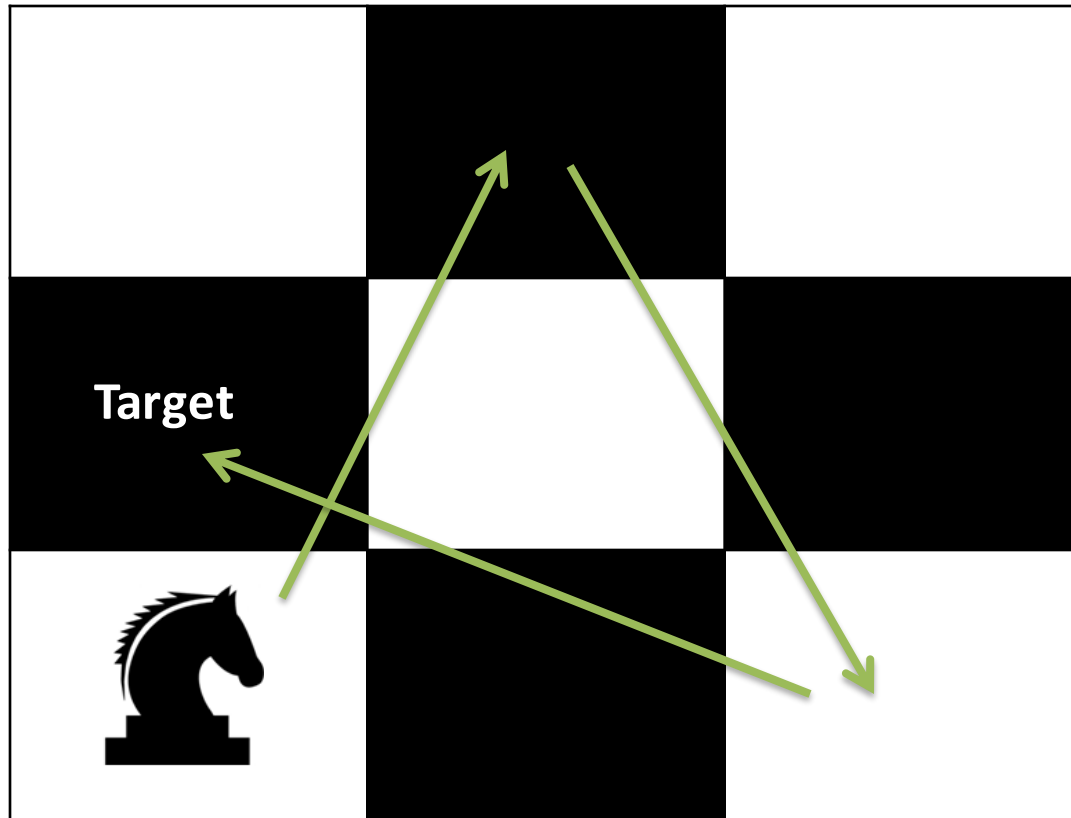
Planning: Knight's Plight

Can we move the given knight to the “Target” location?



Planning: Knight's Plight

Can we move the given knight to the “Target” location?



Calculative Puzzle:

Calculation, Computation, Reasoning

The number of times the digit 0 appears in this puzzle is ?.
The number of times the digit 1 appears in this puzzle is ?.
The number of times the digit 2 appears in this puzzle is ?.
The number of times the digit 3 appears in this puzzle is ?.
The number of times the digit 4 appears in this puzzle is ?.
The number of times the digit 5 appears in this puzzle is ?.
The number of times the digit 6 appears in this puzzle is ?.
The number of times the digit 7 appears in this puzzle is ?.
The number of times the digit 8 appears in this puzzle is ?.
The number of times the digit 9 appears in this puzzle is ?.

Replace ? sign with count of numbers?

Calculative Puzzle:

Calculation, Computation, Reasoning

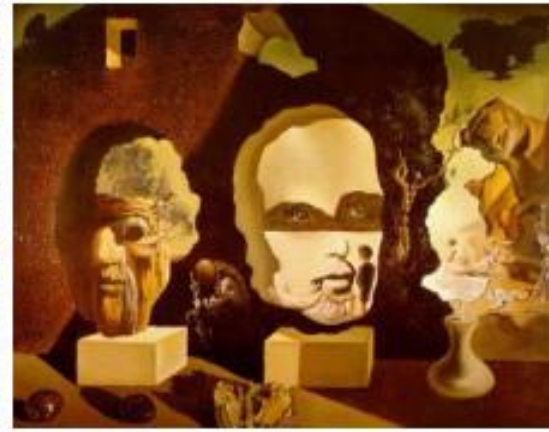
The number of times the digit 0 appears in this puzzle is 1.
The number of times the digit 1 appears in this puzzle is 11.
The number of times the digit 2 appears in this puzzle is 2.
The number of times the digit 3 appears in this puzzle is 1.
The number of times the digit 4 appears in this puzzle is 1.
The number of times the digit 5 appears in this puzzle is 1.
The number of times the digit 6 appears in this puzzle is 1.
The number of times the digit 7 appears in this puzzle is 1.
The number of times the digit 8 appears in this puzzle is 1.
The number of times the digit 9 appears in this puzzle is 1.

Replace ? sign with count of numbers?

Painting by two different painters: Supervised Learning



Painter **A**



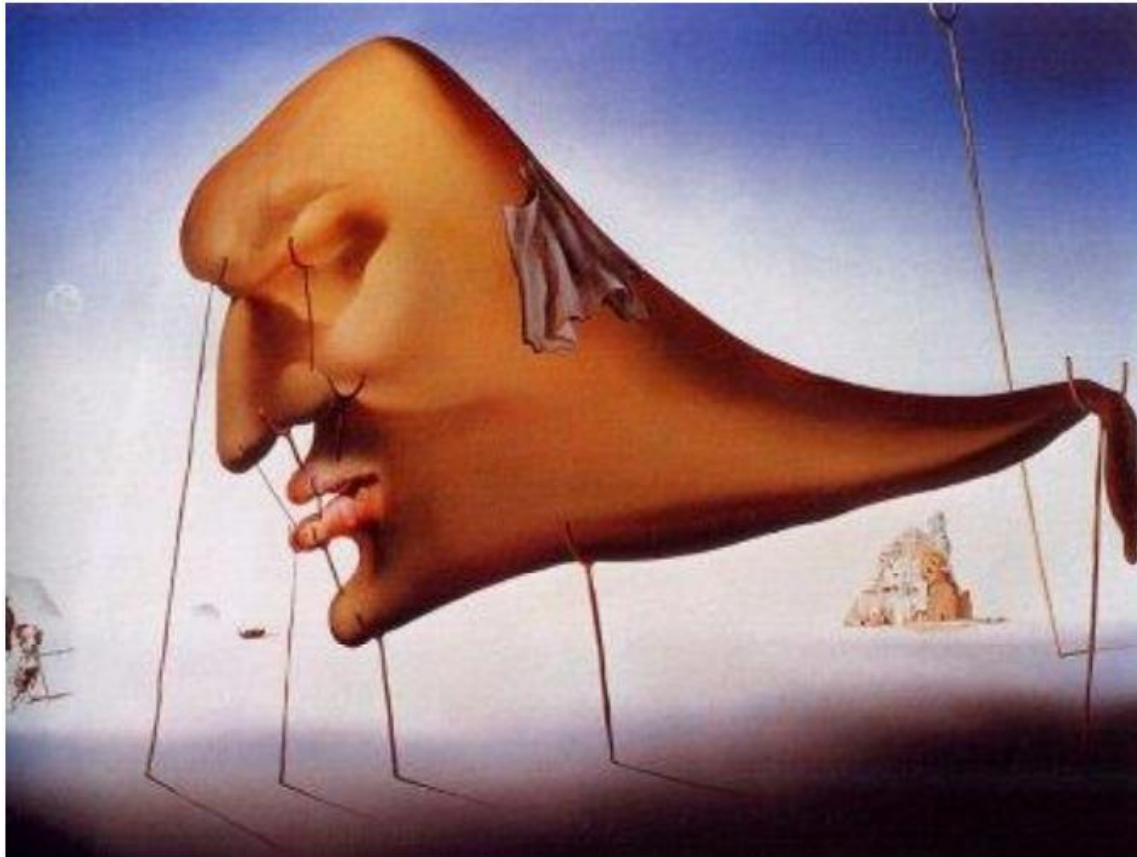
Painter **B**

Now your turn...

Who's painting is this?

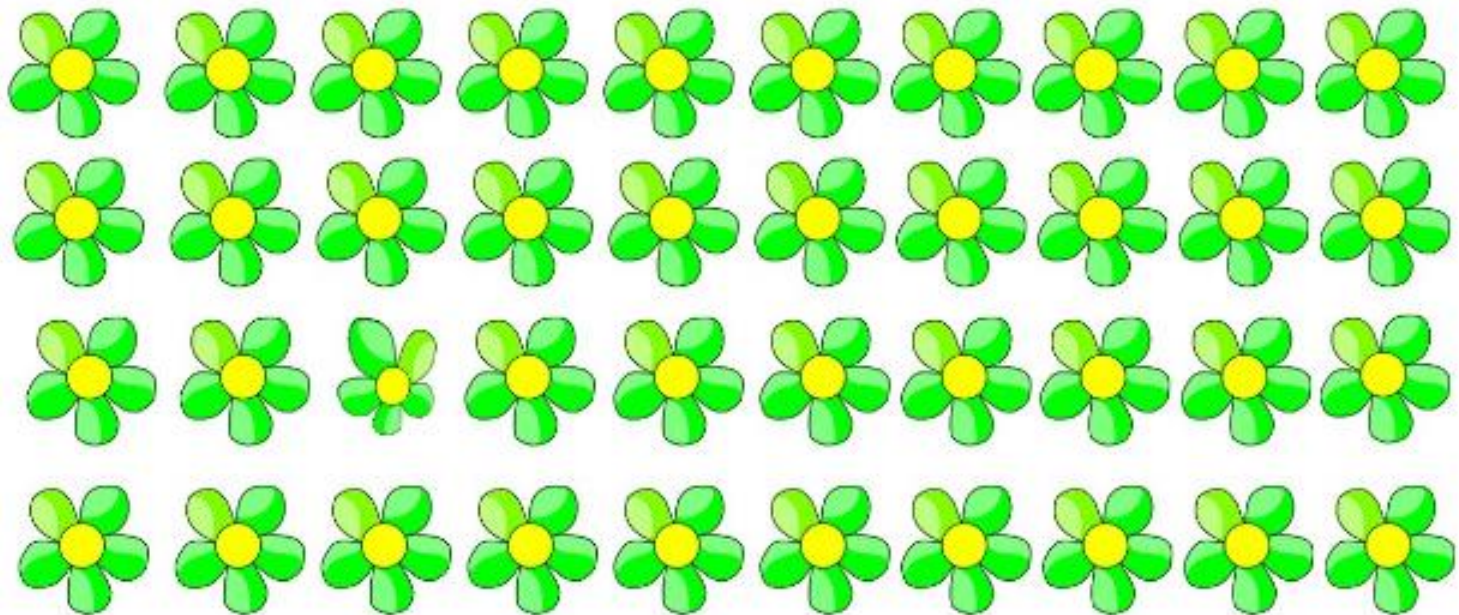
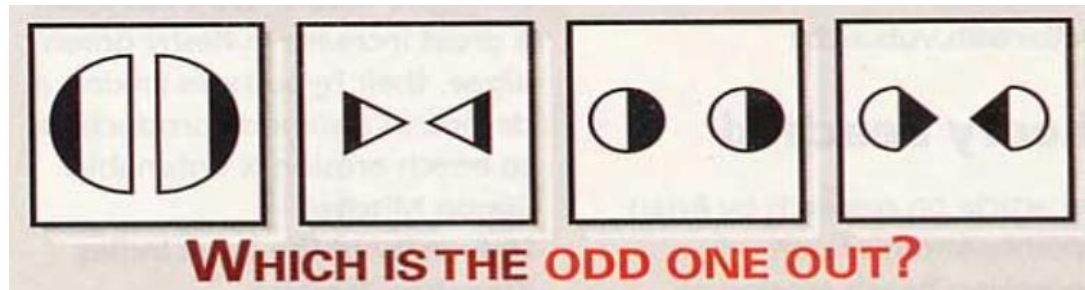


And this?



Finding the odd one out:

Unsupervised Learning

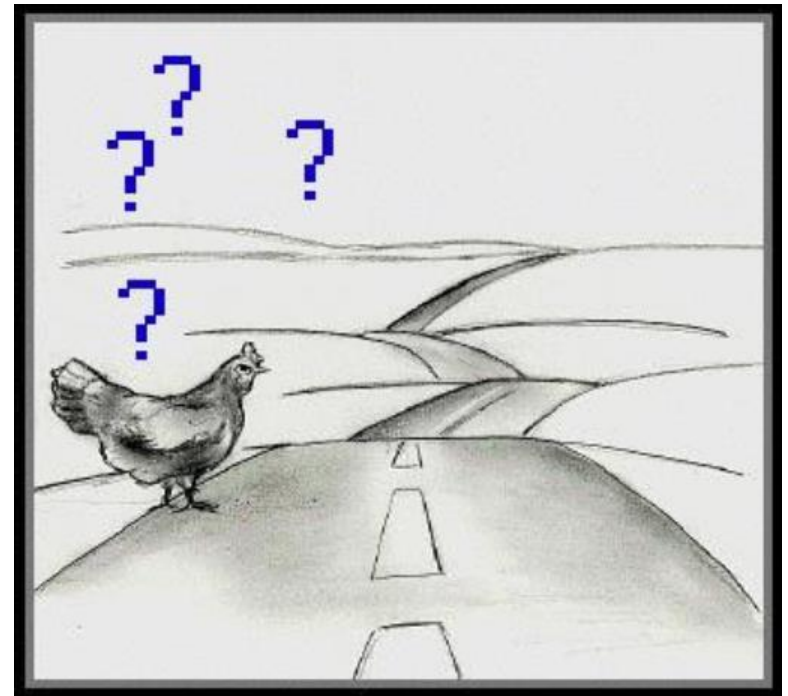


Some more...

What is the next number in this series?

- 1, 1, 2, 3, 5, ?

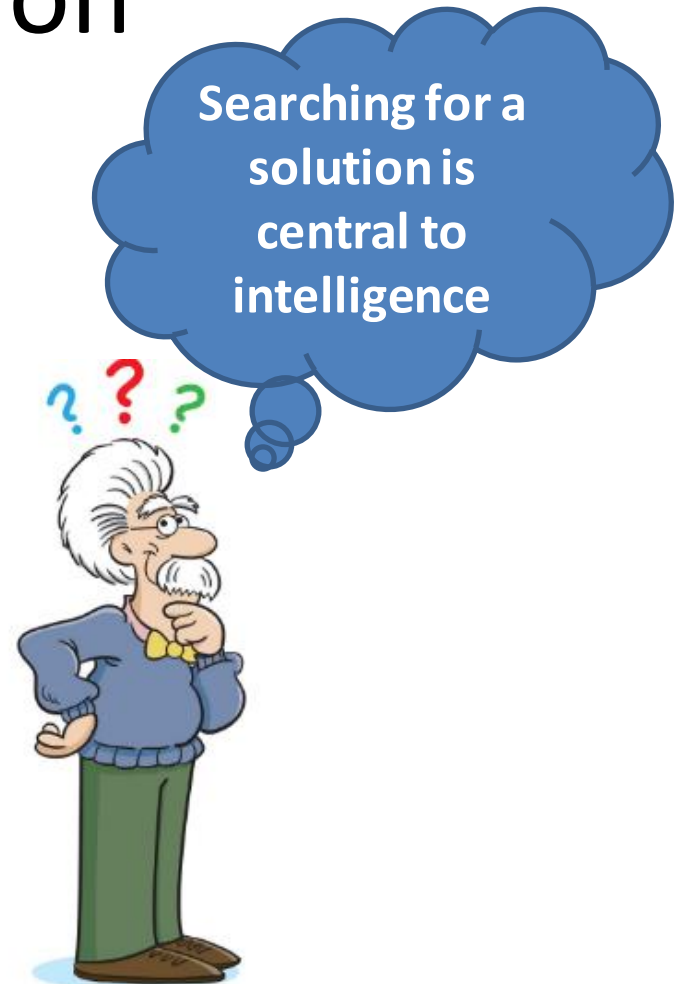
Would you cross a road when a fast car is approaching?



These are the very characteristics of
Intelligent beings!!

Conclusion

- Knight's Plight
 - Planning
- Calculative Puzzle
 - Calculation, Computation, Reasoning
- Finding the odd one out
 - Unsupervised Learning
- Painting by two different painters
 - Supervised Learning
- Series Completion
 - Prediction
- Road Crossing
 - Rational Actions
- Process of Writing this Conclusion
 - Summarizing, Abstraction



What is Artificial Intelligence?

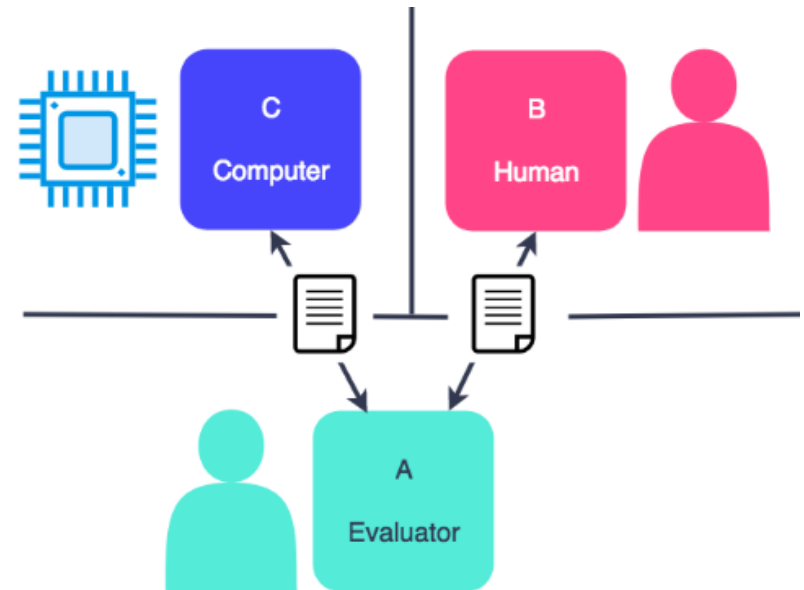
- Computers are Dumb.
- Making a machine (computer) to perform the same tasks which you have just done is called _____.
 - Artificial Intelligence
- If you learn to do these tasks using existing data, then this is called _____.
 - Machine Learning

Artificial Intelligence

- Artificial Intelligence takes the problem of understanding *how we think* a step further
 - It attempts not just to understand it – but – also to build intelligent entities
- A more proper definition of Artificial Intelligence
 - *The art of creating machines that perform functions that require intelligence when performed by people [1]*
- Measurement of Artificial Intelligence
 - Turing Test

Turing Test

- Turing Test
 - Suggested major components of AI: knowledge, reasoning, language understanding, learning
- Application of the Turing Test
 - CAPTCHA: **C**ompletely **A**utomated **P**ublic **T**uring test to tell **C**omputers and **H**umans **A**part



Is AI intelligent?

- Planning
 - Kasparov Vs. IBM Blue (1997)
- Calculation
 - Symbolic Integration in Mathematica
 - Theorem Provers



$$\int e^{2x} \cos 3x \, dx = \frac{1}{3} e^{2x} \sin 3x + \frac{2}{9} e^{2x} \cos 3x - \frac{4}{9} \int e^{2x} \cos 3x \, dx$$
$$+ \frac{4}{9} \int e^{2x} \cos 3x \, dx \qquad + \frac{4}{9} \int e^{2x} \cos 3x \, dx$$

$$\frac{13}{9} \int e^{2x} \cos 3x \, dx = \frac{1}{3} e^{2x} \sin 3x + \frac{2}{9} e^{2x} \cos 3x$$

$$\frac{9}{13} \frac{13}{9} \int e^{2x} \cos 3x \, dx = \frac{9}{13} \left(\frac{1}{3} e^{2x} \sin 3x + \frac{2}{9} e^{2x} \cos 3x \right)$$

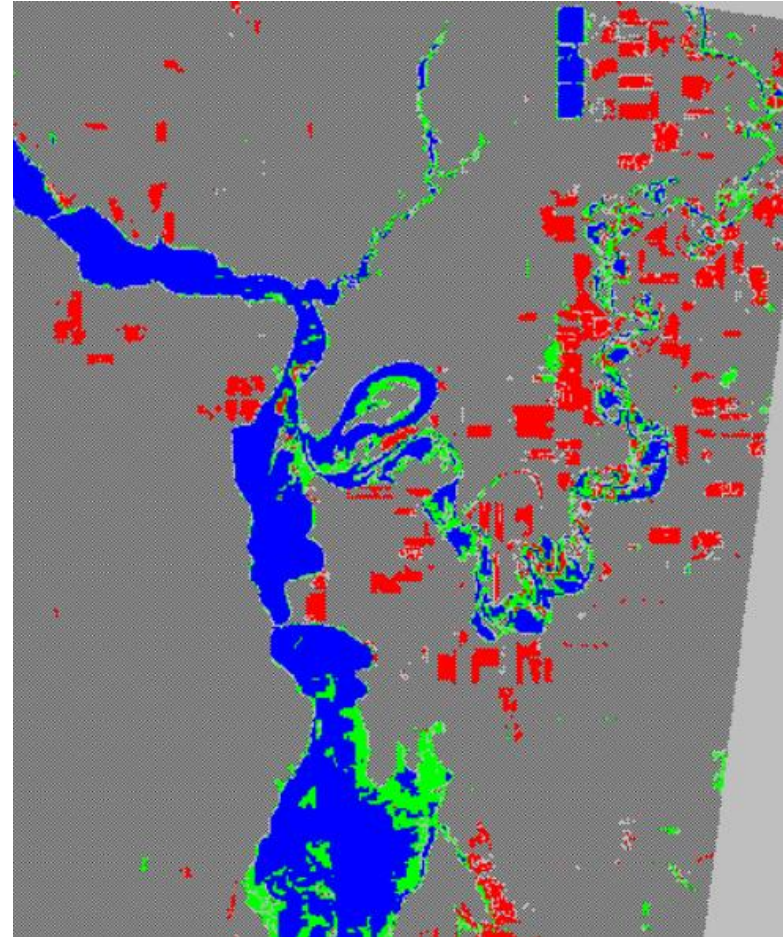
$$\int e^{2x} \cos 3x \, dx = \frac{3}{13} e^{2x} \sin 3x + \frac{2}{13} e^{2x} \cos 3x + C$$

Is AI intelligent?

- Learning without a Teacher
 - ERDAS Imagine
 - Classification of Land Use

ERDAS:

- World Class Remote Sensing Software

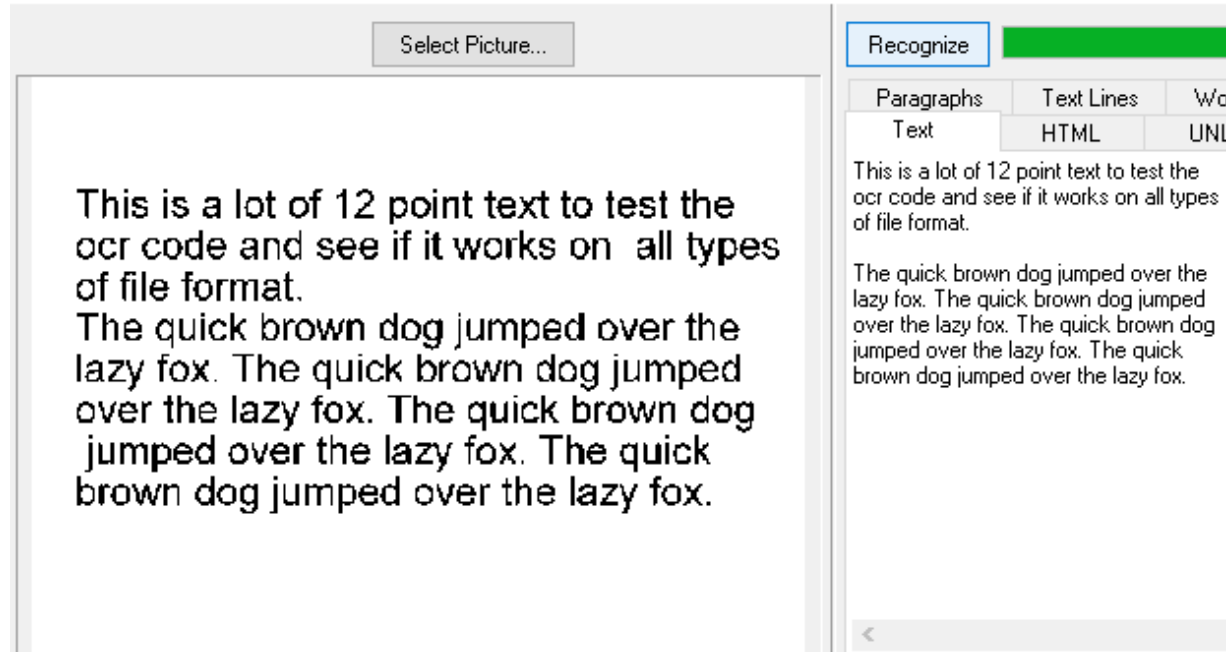


■ = water; ■ = wetland; ■ = marginal habitats; ■ = upland (non-useable habitats).

Is AI intelligent?

OCR Example

- Learning with a teacher
 - No Hands Across America!
 - Optical Character Recognition



Forecasting and Prediction

Block B, Lahore

Monday

Clear with periodic clouds



30°C | °F

Precipitation: 0%

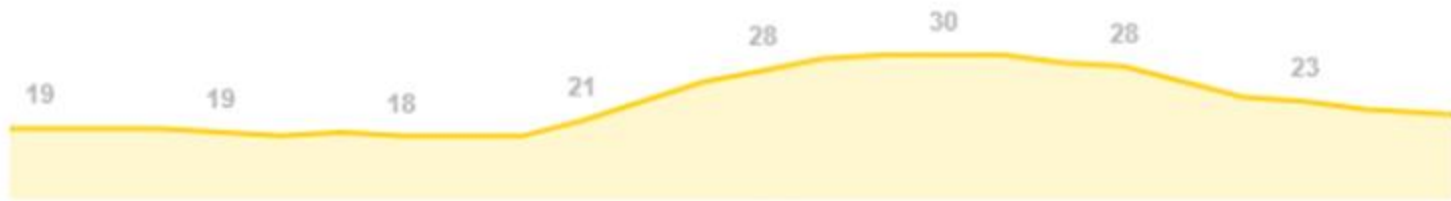
Humidity: 83%

Wind: 0 km/h

Temperature

Precipitation

Wind



12 AM

3 AM

6 AM

9 AM

12 PM

3 PM

6 PM

9 PM

Mon



30° 18°

Tue



31° 17°

Wed



31° 18°

Thu



31° 18°

Fri



27° 16°

Sat



27° 14°

Sun



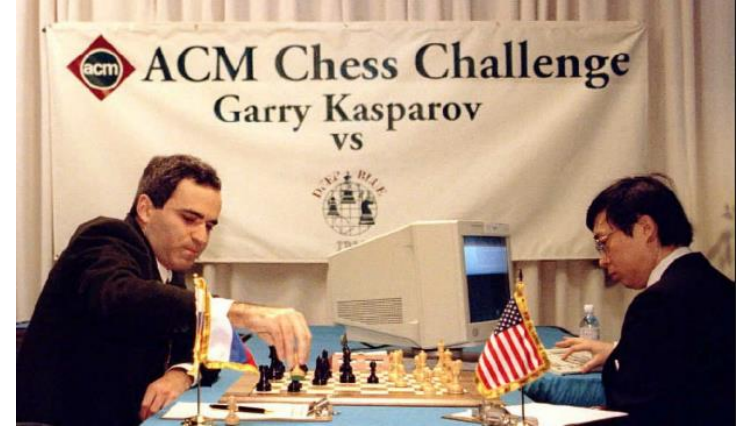
26° 13°

Mon



26° 13°

Is deep blue Intelligent?

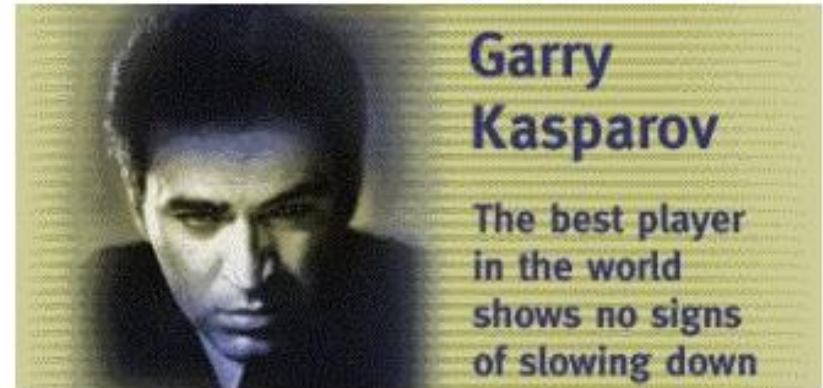


Deep Blue defeated the world chess champion Garry Kasparov.

Does this make Deep Blue **an intelligent machine**?

- Think about it for a few seconds and take a stance
- **Write down at least two items** that support your claim either way
- Pair up with your neighbor and Exchange opinions
- Try to argue for your stance
- Listen carefully to the arguments of your neighbor
- Try to reach an agreement, if possible
- Make your verdict: YES (machine is intelligent) or NO (machine is not intelligent) or X (no agreement)

Deep Blue Vs Garry Kasparov



200,000,000 board configurations
per second

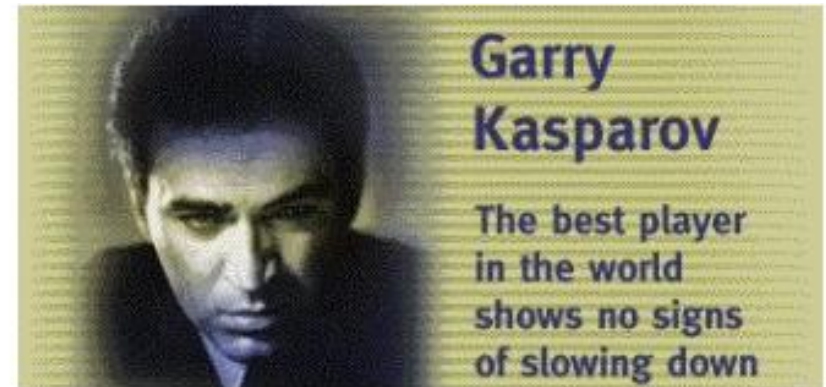
3 board configurations per second

Has **small knowledge** about chess,
but a **huge computational**
capacity

Has **huge knowledge** about chess,
but a considerably **smaller**
computational capacity

A machine **has no emotions nor**
intuition, it does not forget, cannot
be confused or feel uncomfortable

Has feelings and **brilliant intuition**,
but can experience **fatigue and**
boredom and loss of concentration



Deep Blue **does not learn**, therefore it can't use artificial intelligence to learn from its opponent

Deep Blue is incredibly efficient in **solving problems from the domain of chess** but is less "intelligent" even compared to a small child

Garry Kasparov can learn and **adapt quickly** based on his success or failure

Garry Kasparov is **generally very intelligent**: he authored several books and speaks many languages

Natural Language Understanding

- *The man tried to take a picture of a man with a turban.*
 - Did the man try to take a picture with a turban, or
 - take a picture of a man who is wearing a turban?
- *The man saw the boy with the telescope*
- Communicating in natural language assumes **world knowledge and the understanding of context**, both of which are required to resolve the ambiguities

Case Study: Self-driving Car

- Write two pages summary of Autonomous Car in **YOUR OWN WORDS**, take help from google.